INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **BCA** , Semester : **Second**

Stream: **BCA**

Paper Code: **BCACC202**

Paper Name: **Object Oriented Programming with C++**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. What is procedure-oriented programming? What are its main characterictics? **2**

2. What is a class? How does it accomplish data hiding? **2**

3. What are objects? How are they created? **2**

4. How is a member function of a class defined? **2**

5. How do the following statements differ? **2**
   a) $char * const\ p;$
   b) $char\ const * p$

6. Differentiate between single and multiple inheritance. **2**

7. What are seekg( ) and seekp( ) functions in file handling? **2**

8. Show the need of virtual functions? **2**

9. What do you mean by a token? **2**

10. Compare and contrast the variables and constants in C++. What are the rules to be followed for identifiers? **2**

11. Write down the example to overload unary and binary operators in C++. **2**

12. How is an exception handled in C++? **2**

13. When do we used multiple catch handlers? **2**

14. Outline the I/O facilities in C++ that differ from that in C. **2**

15. Compare and contrast error and exception. **2**

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Explain the four pillars of Object-Oriented Programming (OOP) with examples. **5**

17. Write down a C++ program to swap two numbers without using third variable with the help of friend function. **5**

Test Case:

Value before exchange a and b:
100
200
Value after exchange a and b:
200
100

18. Give classification of operators available in C++ with the help of neat and clean diagram. **5**

19. Define ternary operator. Compare it with if and if-else statement. **5**

20. What is the need of type conversion? Discuss different types of type conversion in C++. **5**

21. Write a program that reads a text from the keyboard and displays the following information on the screen in two columns: **5**

(a) Number of lines
(b) Number of words
(c) Number of characters

Strings should be left-justified and numbers should be right-justified in a suitable field width.

22. Describe the various classes available for file operations. **5**

23. Write a program containing a possible exception. Use a try block to through it and a catch block to handle it properly. **5**

24. List the steps involved in using a file in a C++ program? **5**

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. If any array contain huge amount of data (but all the element are in sorted order), it is very difficult task to find out the location of any data item within less amount of time. Hence write down a C++ program to find out the location of any data item with minimum number of comparison by implementing the concept of object and class. At the same time security of the data set is also essential. **10**

Test Case:

Input-1: Element in array: 10,19,27,30,35,67,80
Minimum no. of comparison required: 1
Maximum no. of comparison required: 3
Input-2: Search Item: 19
Output: Location: 2

26. An electricity board charges the following rates to domestic users to discourage large consumption of energy: **10**

- For the first 100 units → 60 Paise per unit
- For the next 200 units → 80 Paise per unit
- Beyond 300 units → 90 Paise per unit

Additionally, all users are charged a Minimum charge of Rs. 50.00. If the total bill exceeds Rs. 300.00, an additional surcharge of 15% is applied. Write a C++ program to read the names of the users and the number of units consumed and print out the charges with names.

27. Let as assume an organization contain n number of employee and the HR wants to sort the salary of their employee and also want to display the salary as per ascending order from manager level to worker level. Write down a C++ program to resolve the problem with the help of class and object. **10**

Hints: If the organization contains 10 employees, the total number of pass will be 9 and the required number of comparison = Element in each pass-1.

28. What is inheritance? Explain the following with examples. **10**

a) Single Inheritance
b) Multiple Inheritance
c) Hybrid Inheritance
d) Hierarchical Inheritance

29. Evaluate a C++ program to store and display the details of student (Roll, Name, Fees) using the concept of constructor, where the number of students will be given by the user. **10**

30. Explain the hierarchy and functionalities of the standard C++ input/output stream classes. Discuss the role of ios, istream, ostream, iostream, and streambuf in handling input and output operations. **10**

31. What are user-defined exceptions? Write a program that defines a custom exception for invalid age input. **10**

32. Write a C++ program to demonstrate the concept of re-throwing an exception. **10**

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